

UNIT 2

Project Scheduling

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

2.1 Experiment 1: Project Evaluation and Review Technique (PERT) Analysis

2.1.1 Work Breakdown Structure (WBS) for the Project

The Work Breakdown Structure decomposes the total project scope into manageable activities across the full 5-month duration from January to May 2026. The project is divided into nine major activities spanning planning, development, testing, deployment, and closure.

Task ID	Activity Description	Optimistic (O)	Most Likely (M)	Pessimistic (P)
A	Requirements Gathering & System Design	5	7	12
B	UI/UX Design & Prototyping (Next.js)	8	12	18
C	Firebase Backend Setup & Authentication	7	10	16
D	AI (Gemini) Integration & Story Module	12	18	28
E	Interactive Abacus & Calculation Modules	15	20	30
F	Number System Explorer & Quiz Module	10	14	22
G	System Integration & Testing	8	12	20
H	Deployment on Vercel & Performance Testing	5	7	12
I	Documentation, User Training & Project Closure	4	5	9

2.1.2 PERT Expected Duration Calculation

The PERT expected duration formula provides a weighted average of the optimistic, most likely, and pessimistic estimates: $T_e = (O + 4M + P) / 6$.

Activity	Calculation	Expected Duration T_e (days)
A	$(5 + 4 \times 7 + 12) / 6$	7.50
B	$(8 + 4 \times 12 + 18) / 6$	12.33
C	$(7 + 4 \times 10 + 16) / 6$	10.50
D	$(12 + 4 \times 18 + 28) / 6$	18.67
E	$(15 + 4 \times 20 + 30) / 6$	20.83
F	$(10 + 4 \times 14 + 22) / 6$	14.67
G	$(8 + 4 \times 12 + 20) / 6$	12.67
H	$(5 + 4 \times 7 + 12) / 6$	7.50
I	$(4 + 4 \times 5 + 9) / 6$	5.50

2.1.3 Variance and Standard Deviation

Variance and standard deviation measure the uncertainty in each activity duration. Variance is calculated using $((P - O) / 6)^2$, and standard deviation is the square root of variance.

Activity	Calculation	Variance (σ^2)	Std Dev (σ)
A	$((12-5)/6)^2$	1.36	1.17
B	$((18-8)/6)^2$	2.78	1.67
C	$((16-7)/6)^2$	2.25	1.50
D	$((28-12)/6)^2$	7.11	2.67
E	$((30-15)/6)^2$	6.25	2.50
F	$((22-10)/6)^2$	4.00	2.00
G	$((20-8)/6)^2$	4.00	2.00
H	$((12-5)/6)^2$	1.36	1.17
I	$((9-4)/6)^2$	0.69	0.83

2.1.4 Probability Analysis

The total variance along the critical path $A \rightarrow B \rightarrow D \rightarrow E \rightarrow G \rightarrow H \rightarrow I$ is 23.55, giving a total standard deviation of 4.85 days. For a target completion within 95 days, the Z-score is 2.06, which corresponds to a probability of approximately 98.03%. This indicates a very high confidence of completing the project within 95 days.

2.2 Experiment 2: Critical Path Method (CPM) Analysis

2.2.1 Precedence Relationships

Activity	Predecessor(s)	Rationale
A	—	Starting activity; requirements must be finalized first
B	A	UI/UX design requires approved requirements
C	A	Firebase setup requires defined data models from requirements
D	B, C	Gemini AI module needs UI framework and backend ready
E	D	Abacus module requires AI module and backend integration complete
F	D	Number explorer and quiz modules require AI/backend integration
G	E, F	Integration/testing require all feature modules complete
H	G	Deployment requires successful integration testing
I	H	Documentation and closure require successful deployment

2.2.2 Forward Pass (Earliest Start & Earliest Finish)

Activity	Earliest Start (ES)	Expected Duration (Te)	Earliest Finish (EF)
A	0	7.50	7.50
B	7.50	12.33	19.83
C	7.50	10.50	18.00
D	19.83	18.67	38.50
E	38.50	20.83	59.33
F	38.50	14.67	53.17
G	59.33	12.67	72.00
H	72.00	7.50	79.50
I	79.50	5.50	85.00

The total project duration from the forward pass is 85 days, approximately 17 weeks or 4.25 months.

2.2.3 Backward Pass and Slack Calculation

Activity	Latest Start (LS)	Latest Finish (LF)	Slack (LS – ES)	Critical?
A	0.00	7.50	0.00	Yes
B	7.50	19.83	0.00	Yes
C	9.33	19.83	1.83	No
D	19.83	38.50	0.00	Yes
E	38.50	59.33	0.00	Yes
F	44.67	59.33	6.17	No
G	59.33	72.00	0.00	Yes
H	72.00	79.50	0.00	Yes
I	79.50	85.00	0.00	Yes

2.2.4 Critical Path Identification

The critical path is A → B → D → E → G → H → I, with a total duration of 85 days. Activities C and F are non-critical and have 1.83 and 6.17 days of float respectively.

2.2.5 Gantt Chart Representation

Activity	Duration (days)	Timeline
A: Requirements	7.5	Day 0 – Day 8
B: UI/UX Design	12.33	Day 8 – Day 20
C: Firebase Setup	10.5	Day 8 – Day 18
D: AI/Gemini Module	18.67	Day 20 – Day 39
E: Abacus & Calc Modules	20.83	Day 39 – Day 59

Activity	Duration (days)	Timeline
F: Number Explorer & Quiz	14.67	Day 39 – Day 53
G: Integration & Testing	12.67	Day 59 – Day 72
H: Deployment & Perf Test	7.5	Day 72 – Day 80
I: Docs & Closure	5.5	Day 80 – Day 85

2.3 Experiment 3: Project Resource Levelling Techniques

2.3.1 Project Team Formation and Structure

Role	Name	Key Responsibilities
Project Manager	Ashray Singh	Planning, scheduling, risk management, stakeholder communication
Full-Stack Developer	Madhav Nehra	Next.js frontend, Firebase backend, Vercel deployment
AI & Content Engineer	Rudransh Yadav	Gemini integration, educational content, quiz and story modules

2.3.2 Selection of Project Manager

Ashray Singh was selected as Project Manager based on strong understanding of the curriculum, prior experience with software planning and documentation, and the ability to coordinate technical and academic deliverables.

2.3.3 Roles and Responsibilities of Project Manager

Area	Responsibilities
Planning	Define WBS, create PERT/CPM network, establish baseline schedule
Execution	Coordinate sprint activities, assign tasks, manage workload
Monitoring	Track progress, EVA, schedule and cost variances
Communication	Faculty updates, daily stand-ups, milestone reviews
Quality	Coordinate testing and ensure deliverable quality
Risk Management	Identify and mitigate project risks
Closure	Final report preparation and lessons learned

2.3.4 Resource Requirements by Activity

Activity	Duration	Developer	AI Engineer	PM	Total Resources
A	7.50	1	0	1	2
B	12.33	1	1	0	2
C	10.50	1	0	0	1

Activity	Duration	Developer	AI Engineer	PM	Total Resources
D	18.67	1	1	0	2
E	20.83	1	1	0	2
F	14.67	1	1	0	2
G	12.67	1	1	1	3
H	7.50	1	0	1	2
I	5.50	0	1	1	2

2.3.5 Resource Profile Before Levelling

Time Period	Activities	Developer Load	Constraint Violation
Day 0–8	A	1	None
Day 8–20	B + C	2	Overload — only 1 developer available
Day 20–39	D	1	None
Day 39–53	E + F	1 (split)	Minor — manageable
Day 53–59	E only	1	None
Day 59–72	G	1	None
Day 72–80	H	1	None
Day 80–85	I	0	None

2.3.6 Resource Levelling Strategy

Resource levelling resolved the developer overload during Days 8–20 by delaying Activity C within its float and prioritizing the most intensive design work in Activity B. Additional levelling was achieved by splitting work between the developer and AI engineer during Activities E and F, and by assigning non-coding coordination work to the Project Manager.

2.3.7 Resource Profile After Levelling

Time Period	Developer Load	Status
Day 0–8	1	OK
Day 8–14	1 (B only)	OK
Day 14–20	1 (B + C sequential)	OK
Day 20–39	1 (D)	OK
Day 39–53	1 (E; F handled by AI Engineer)	OK
Day 53–72	1	OK
Day 72–80	1	OK
Day 80–85	0	OK

2.3.8 Impact of Resource Levelling

Parameter	Before Levelling	After Levelling
Peak Developer Load	2	1
Constraint Violations	1 period	0 periods
Project Duration	85 days	86.83 days
Schedule Increase	—	~2 days
Additional Cost	—	INR 0

2.4 Experiment 4: Project Crashing Techniques

2.4.1 Crash Analysis Rationale

Project crashing is used when the project must be completed earlier than the baseline duration. Only critical path activities are candidates for crashing because reducing non-critical activities does not reduce the overall project completion time.

2.4.2 Activity Crash Options

Activity	Normal Duration	Crash Duration	Crash Cost/Day (INR)	Critical?
B (UI/UX Design)	12.33	9	3000	Yes
D (AI/Gemini Module)	18.67	14	5000	Yes
E (Abacus & Calc)	20.83	17	4000	Yes
G (Integration & Testing)	12.67	10	3500	Yes
H (Deployment)	7.50	6	2500	Yes

2.4.3 Least Cost Crashing

Step	Action	Days Crashed	Cumulative Crash	Cost (INR)
1	Crash H	2.5	2.5	6250
2	Crash B	3	5.5	9000
3	Crash G	2.5	8	8750
4	Crash E	2	10	8000
Total	—	10	10	32000

2.4.4 Crash Summary

Parameter	Normal Schedule	Crashed Schedule
Project Duration	85 days	75 days
Total Crash Cost	0	INR 32,000
Schedule Reduction	0	10 days
Average Cost per Day Saved	—	INR 3,200
Within Contingency?	—	Yes

2.4.5 Time-Cost Trade-off Summary

Duration (days)	Total Crash Cost (INR)	Marginal Cost (INR/day)	Feasibility
85	0	—	Baseline
82.5	6250	2500	Recommended
79.5	15250	3000	Feasible
77.0	24000	3500	Possible
75.0	32000	4000	Maximum recommended
73.0	40000	4000	Not recommended

2.4.6 Conclusion for Unit 2

Unit 2 established the scheduling framework for the project through PERT, CPM, resource levelling, and project crashing. The critical path was identified as $A \rightarrow B \rightarrow D \rightarrow E \rightarrow G \rightarrow H \rightarrow I$ with a baseline duration of 85 days. Resource levelling resolved the developer overallocation problem with no additional cost, while project crashing demonstrated that the schedule could be compressed to 75 days at an additional cost of INR 32,000 if needed.

Unit 2 Figures

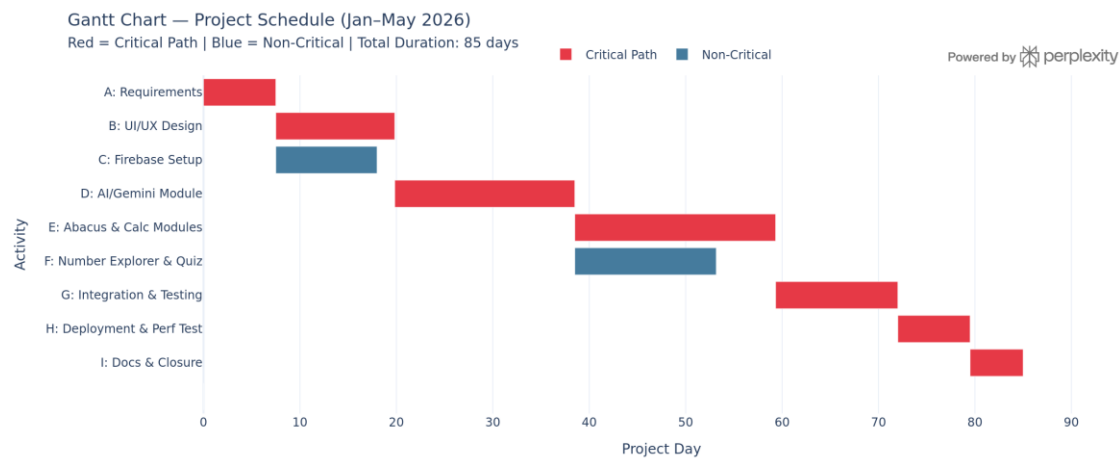


Figure: Gantt Chart



Figure: Resource Levelling Profile